

Abhinav Kumar Chaudhary

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience in distributed **LLM inference**, **RAG**, and **asynchronous backend engineering**, with end-to-end ownership from model integration to production cloud deployment. Skilled in **multimodal AI pipelines**, **document intelligence**, and **multilingual NLP**, with a track record of shipping measurable, recruiter-verifiable impact across AI/ML projects.

WORK EXPERIENCE

ScholarRank AI

AI/ML Engineer Intern

Remote

Mar 2025 – Aug 2025

- Improved ML model accuracy by 15% using XGBoost feature engineering, hyperparameter tuning (grid/random search), and class-imbalance handling on a 50K+ record production dataset.
- Reduced API latency by 30% by architecting a FastAPI based inference microservice enabling real-time model serving at 100+ requests/min throughput.
- Delivered production-ready ML pipelines end-to-end within an Agile team – covering data preprocessing, feature engineering, model training, and CI/CD-driven deployment.

PROJECTS

SynthexAI — Multi-LLM Orchestration & Agentic Inference Engine

[GitHub](#) | [Live Demo](#)

Stack: Python, FastAPI, asyncio, Groq, Gemini Flash, Mistral, Tavily, AWS EC2, Nginx, Unicorn, Systemd

- Engineered a multi-LLM orchestration engine that queries **Groq (Llama 3 70B)**, **Google Gemini 1.5 Flash**, and **Mistral Small** concurrently via **Python asyncio**, applying outlier-response filtering and lead-model synthesis to merge three independent completions into one grounded answer – cutting response latency by 67% versus sequential API calls.
- Architected a multimodal, safety-aware input layer – **Groq Whisper Large V3** for voice-to-text, **Groq Vision (Llama 4 Scout)** for **OCR**, **Tavily API** for cited real-time web search – guarded by pre-query harmful-content filtering and a cross-model confidence score surfaced to the user for transparency.
- Deployed the full system on **AWS EC2** behind **Nginx** (reverse proxy) and **Unicorn**, supervised by **Systemd** for persistent uptime, with **Server-Sent Events (SSE)** streaming for real-time token-by-token response delivery to a custom floating-widget frontend.

LexiMind AI — Dual-Mode Retrieval-Augmented Generation Pipeline

[GitHub](#) | [Demo](#)

Stack: Python, LangChain, FAISS, HuggingFace (all-MiniLM-L6-v2), FastAPI

- Developed a dual-mode chatbot system that automatically switches between general LLM conversation and **RAG-based document Q&A**, using **FAISS** vector search and **HuggingFace Sentence-Transformer (all-MiniLM-L6-v2)** embeddings to ground answers strictly in uploaded PDF content.
- Implemented stateful conversational memory (last 10 messages) with a dedicated **/clear** reset endpoint, and designed a strict **hallucination-control** system prompt that forces the model to answer only from retrieved context or explicitly say "I don't know."

- Progressively refined the system across 4 architectural stages (stateless → stateful → streaming → RAG) before integrating them into one **FastAPI** backend, independently debugging real production issues including **FAISS**-NumPy version conflicts and vector-DB state leaks across uploads.

TECHNICAL SKILLS

Programming & Data Analysis: Python, SQL, MySQL, Pandas, NumPy, Statistical Analysis, A/B Testing, Matplotlib, Seaborn

Generative AI & LLMs: RAG, LangChain, AI Agents, Prompt Engineering, Semantic Search, OpenAI/Gemini/Claude/Groq APIs, FAISS, LoRA, QLoRA, PEFT, Quantization, Instruction Tuning

Machine Learning & NLP: Scikit-learn, XGBoost, Classification, Regression, Feature Engineering, Hyperparameter Tuning, SHAP, OCR, Embeddings, Text Classification, Information Extraction, Multilingual Processing

Backend & Deployment: FastAPI, Flask, REST APIs, Docker, AWS EC2, Nginx, Unicorn

Tools & Automation: Git, MLflow, Jupyter Notebook, VS Code, Linux, n8n, Make

EDUCATION

Noida Institute of Engineering and Technology (NIET), Greater Noida

B.Tech – Information Technology

Aug 2022 – May 2026

CERTIFICATIONS

AWS AI Practitioner Challenge – Agentic AI & Cloud AI | Amazon Web Services (AWS)

Natural Language Processing using Python | Infosys Springboard

Python for Data Science, AI & Development | IBM / Coursera

Introduction to AI & ML | Infosys Springboard

LEADERSHIP

- **Head Coordinator**, Green Gold Society (NIET) – increased student engagement by 30%+ across 1,000+ students.
- **Team Lead** – FormSathi AI Project – led a 5-member Agile team to build an AI document understanding system (OCR + Multilingual LLM + RAG pipeline), delivered on schedule with a 70%+ reduction in manual document-processing effort.